

National University of Computer and Emerging Sciences

Semester: Fall 2015

Course: Applied Calculus (EE) Mid: 2

Time: 1 hr

Total Marks: 30

Question.1: Find the critical points of

$$f(x) = x^{1/3}(x - 4)$$

Identify the intervals on which f is increasing and decreasing. Find the function's local and absolute extreme values.

Question.2: Evaluate the following integral, if exist and give reason if does not exist.

$$\int_{-1}^2 (x - 2|x|)dx$$

Question.3: Sketch the region enclosed by the given curves. Describe whether to integrate with respect to x or y (with reason). Then find the area of the region.

$$y = 8 - x^2, \quad y = x^2, \quad x = -3, \quad x = 3$$